In **T 632/91** the board stated that evidence which did not comprise a comparison of the claimed subject-matter with the state of the art might nevertheless rebut a prima facie assumption that there existed some common general knowledge which would have allowed the skilled person to disregard structural differences in chemical compounds.

In **T 984/15** the board recalled that the technical field and general knowledge associated with the notional skilled person within the meaning of Art. 56 EPC should be defined on the basis of the objective technical problem to be solved by that skilled person in the framework of the problem-solution approach. This was because the skilled person under Art. 56 EPC is the person qualified to solve the established objective technical problem (see e.g. **T 32/81**, OJ 1982, 225; **T 422/93**, OJ 1997, 25; **T 1140/09**; **T 1523/11**; **T 1450/16**).

In **T 1601/15** the board held that the skilled person did not need prompting to apply their knowledge. It was not convinced by the argument that the skilled person would have had no reason to draw upon it. The skilled person did not need a reason to apply their knowledge. This knowledge formed, as it were, the technical background for any activities they performed and fed into all their decisions. In this regard, a distinction had to be made between the common general knowledge in the art and the teaching of specialist documents.

In **T 206/83** (OJ 1987, 5) the board pointed out that normally patent specifications were not part of common general knowledge. In **T 1540/14** the board held that the common general knowledge of the skilled person was not normally established on the basis of the content of patent documents (see also e.g. **T 671/94**).

It has been emphasised in several decision that the assertion that something was part of the common general knowledge needed only to be substantiated if challenged by another party or the EPO (see e.g. **T 766/91**, **T 234/93** . Where an assertion that something was part of the common general knowledge is challenged, the person making the assertion must provide proof that the alleged subject-matter indeed forms part of the common general knowledge (**T 766/91**; **T 939/92**, OJ 1996, 309; **T 329/04**; **T 941/04**; **T 690/06**).

8.4. Everyday items from a different technical field

In **T 1043/98** the patent concerned an inflatable gas-bag for a vehicle restraint system, one part being club-shaped and the other generally butterfly-shaped. According to the appellant, the skilled person would immediately arrive at the claimed gas-bag from his knowledge of tennis-ball or baseball construction. This raised the issue of the application of features or solutions drawn from another technical field but which could be considered "everyday items".

In **T 397/87** the board had already pointed out that there was no obvious reason why a skilled person trying to solve a non-trivial problem should have been led to the claimed process by simple examples from everyday life which were unrelated to the problem in question. In **T 349/96**, too, the board was unable to see why the fact that different transport containers were used for beer bottles in an everyday context should prompt a skilled person to invent a spinning/winding machine combination with an integrated transport system even if the many citations from the relevant technical field were unable to do this (see also **T 234/91**).

In **T 234/96**, however, the board concurred with the examining division's view that the skilled person dealing with the practicalities of motorising a dispenser drawer for washing powder had in mind as a model the disc tray of a CD player with push-button electromotor operation, which, at the time of filing the application, was familiar to anyone and which therefore suggested the subject-matter of claim 1. In the board's view, the fact that washing machines and CD players were intrinsically different items serving different purposes did not suffice to prevent the skilled person concerned with the construction of washing machines from taking into consideration the basic principle of automatic tray operation in CD players when designing a dispenser drawer for washing powder.

From a comparison of the above-mentioned decisions, the board in **T 1043/98** concluded that the relevance of such items for inventive step depended very much on the circumstances of the individual case. It agreed that persons skilled in developing the gas-bags in question would include tennis or baseball players. It could not, however, share the appellant's view that to solve the problem addressed by the invention the skilled person would draw on what he might know about tennis-ball or baseball construction. The main reason was that the gas-bag was not intended to be spherical in shape. It was therefore unlikely that the skilled person would take as his starting point an object which was the epitome of a sphere (see **T 477/96**, where the board also concluded that everyday experience was not relevant to the technical field of the invention).

9. Assessment of inventive step

9.1. Assessment of inventive step in the case of mixed-type inventions

The case law of the boards of appeal on the assessment of inventive step in cases where the invention consisted of a mix of technical and non-technical features is as follows: It is legitimate to have a mix of technical and "non-technical" features appearing in a claim, in which the non-technical features may even form a dominating part of the claimed subject matter. Novelty and inventive step, however, can be based only on technical features, which thus have to be clearly defined in the claim (**T 641/00**, OJ 2003, 352; **T 154/04**, OJ 2008, 46; **G 3/08**, OJ 2011, 10; **T 116/06**; **T 1769/10**; **T 209/91**; as similarly held in the earlier **T 26/86**).

Inventive step can be based only on technical features. Non-technical features, to the extent that they do not interact with the technical subject-matter of the claim for solving a technical problem, i.e. non-technical features "as such", do not provide a technical contribution to the prior art and are thus ignored in assessing novelty and inventive step (**T 641/00**, OJ 2003, 352; **T 154/04**, OJ 2008, 46). When assessing the inventive step of such a mixed-type invention, all those features which contribute to the technical character of the invention are taken into account. These also include the features which, when taken in isolation, are non-technical, but do, in the context of the invention, contribute to producing a technical effect serving a technical purpose, thereby contributing to the technical character of the invention. However, features which do not contribute to the technical character of the invention cannot support the presence of an inventive step (**T 641/00**, OJ 2003, 352).

9.2. Problem-solution approach when applied to mixed-type inventions

For the purpose of the problem-solution approach, the problem must be a technical problem which the skilled person in the particular technical field might be asked to solve at the relevant priority date. The technical problem may be formulated using an aim to be achieved in a non-technical field, and which is thus not part of the technical contribution provided by the invention to the prior art. This may be done in particular to define a constraint that has to be met (even if the aim stems from an a posteriori knowledge of the invention) (**T 641/00**; **T 172/03**; **T 154/04**, OJ 2008, 46; **T 116/06**; **G 3/08**, OJ 2011, 10; **T 1769/10**).

9.2.1 The Comvik approach

The Comvik approach was applied in **T 641/00** (OJ 2003, 352). It replaced the "contribution approach" customary until then (for the contribution approach, see in particular **T 121/85**; **T 38/86**, OJ 1990, 384; **T 95/86**; **T 603/89**, OJ 1992, 230; **T 71/91**; **T 236/91**; **T 833/91**; **T 77/92**). The contribution approach had been abandoned as early as **T 1173/97** (OJ 1999, 609, point 8 of the Reasons; see also **G 3/08**, OJ 2011, 10, point 10.6 of the Reasons), it being held that features already known from the prior art could form the basis for a finding that the requirements of Art. 52 (2) and (3) EPC were met. The Comvik approach (**T 641/00**) is a conventional application of the problem-solution approach where the differences from the closest prior art are determined and only those that contribute to technical character are considered for inventive step.

In **T 641/00** the board specified that an inventive step can only be acknowledged for an invention which makes a technical contribution to the art (see, inter alia, **T 38/86**, OJ 1990, 384; **T 1173/97**; **T 1784/06**, **T 258/03** (OJ 2004, 575)), i.e. a contribution in a technical field. The case law of the boards of appeal has also developed the understanding that the examples of subject-matter excluded from patentability given in Art. 52(2) and (3) EPC 1973 concern fields considered to be non-technical.

In **T 258/03** (OJ 2004, 575) the board noted that one compelling reason for not refusing subject-matter consisting of technical and non-technical features under Art. 52(2) EPC 1973 is simply that the technical features may in themselves turn out to fulfil all the requirements of Art. 52(1) EPC 1973. It is often difficult to separate a claim into technical and non-technical features, and an invention may have technical aspects which are hidden in a largely non-technical context. Such technical aspects may be easier to identify within the framework of the examination as to inventive step, which, in accordance with the jurisprudence of the boards of appeal, is concerned with the technical aspects of an invention. Thus, in addition to the restrictive wording of Art. 52(3) EPC 1973 limiting the applicability of Art. 52(2) EPC 1973, there may be practical reasons for generally regarding mixes of technical and non-technical features as inventions within the meaning of Art. 52(1) EPC 1973.

In **T 531/03** the board stated that an attempt to take into account the contribution of non-technical and technical aspects on an equal footing in the assessment of inventive step would not be in conformity with the EPC, since the presence of inventive step would in such an approach be attributed to features which are defined in the EPC as not being an invention.

In **T 912/05**, although the board agreed with **T 641/00** (OJ 2003, 352) and **T 258/03** (OJ 2004, 575), it found that in the case at issue it was not necessary to seek to separate features that were essentially business-related, and thus not relevant for the solution of a technical problem, from those features that, as essentially technical, should be taken into account when assessing inventive step. It concluded that the assessment of the inventive step of a business-related method might be possible without a preliminary clear-cut separation between business-related features and technical features.

In **T 1462/11** the board held that, as has been the case since **T 641/00**, non-technical elements do not contribute to inventive step, and, to that end, may appear in the formulation of the objective technical problem. If the essential idea of the invention lies in a non-technical field (usually one excluded by Art. 52(2) EPC), the objective technical problem often amounts to a statement of requirements that any implementation must meet.

In **T 1784/06** the board noted that the established way of assessing inventive step under Art. 56 EPC made reference to Art. 52 EPC. However, this did not mean that the assessment of the two requirements was "mixed". In **T 1461/12** as well, the board observed that the Comvik approach did not involve an inadmissible mixing of the requirements of Art. 52 and 56 EPC, which were still to be examined separately. Inventive step, however, required a technical contribution to the prior art (see, in particular, **T 38/86**, OJ 1990, 384; **T 1173/97**; **T 1784/06**), i.e. a contribution in a technical field.

9.2.2 Excluded subject-matter

It is the established case law of the boards of appeal (see **G 3/08**, OJ 2011, 10; **T 258/03**, and **T 424/03**, **T 313/10**) that claimed subject-matter specifying at least one feature not falling within the ambit of Art. 52(2) EPC is not excluded from patentability by the provisions of Art. 52(2) and (3) EPC. Features which would, taken in isolation, belong to the matters excluded from patentability by Art. 52(2) EPC may nonetheless contribute to the technical character of a claimed invention, and therefore cannot be discarded in the consideration of the inventive step. This principle was already laid down, albeit in the context of the so-called "contribution approach", in one of the earliest decisions of the boards of appeal to deal with Art. 52(2) EPC, namely **T 208/84** (point 4 ff. of the Reasons) (see also **T 1784/06**).

The presence of an inventive step can only be established on the basis of the technical aspects of both the distinguishing features of, and the effects achieved by the claimed invention over the closest state of the art (**T 641/00**, OJ 2003, 352) (see also **T 619/02**, OJ 2007, 63).

In **T 1543/06** the board stated that the principle as expressed in **T 641/00** (OJ 2003, 352) could be also reformulated as follows: an invention which **as a whole** falls outside the exclusion zone of Art. 52(2) EPC 1973 (i.e. is technical in character) cannot rely on excluded subject matter alone, even if novel and non-obvious (in the colloquial sense of the word), for it to be considered to meet the requirement of inventive step (see also **T 336/07**).

The presence of an inventive step cannot rely on excluded (non-technical) subject matter alone, however original it may be (see also **T 336/07**, see summary in this chapter I.D.9.2.14).

9.2.3 Technical implementation of excluded subject-matter

In **T 1543/06** the board noted that a **mere technical implementation** of excluded subject-matter could not per se form the basis for inventive step (see also **T 1793/07**, **T 337/06**, **T 1225/10** in chapter I.D.9.2.14). Inventive step can be based only on the particular manner of implementation. (**T 859/07**, **T 414/12**).

In the context of the problem-solution approach this can be rephrased as a fictional technical problem in which the per se excluded subject matter appears as an aim to be achieved, see **T 641/00** (Headnote II). Where such excluded subject matter is novel such a formulation of the problem seemingly implies that such matter may be regarded as a given in the assessment of inventive step, which thus appears to depart from what is in fact a hidden starting point. The board viewed this fiction as an artefact of the systematic use of the problem-solution approach for assessing inventive step and the need to differentiate between excluded and non-excluded matter. This should not detract from the basic tenet that excluded subject-matter cannot form the sole basis for a patentable invention. A consideration of the particular manner of implementation must focus on **any further technical advantages** or effects associated with the specific features of implementation over and above the effects and advantages inherent in the excluded subject-matter (**T 336/07**). The latter are at best to be regarded as incidental to that implementation.

The board went on to state that the explicit requirement of a **"further" technical effect** was first formulated for computer-related inventions in decisions **T 1173/97** (OJ 1999, 609) (see also **T 935/97**), but the same principle holds also for other categories of excluded subject-matter which may inherently possess some "technical" effect. In fact, inherent and arguably technical effects may be easily identified for practically all excluded subject-matter, for example such a simple one as reducing time when using or performing it. This is why it needs to be stressed that the **"further" technical effect** cannot be the same one which is **inherent** in the excluded subject-matter itself (**T 2449/10**, **T 1225/10**, **T 1547/09**, **T 1782/09**, **T 2127/09**, **T 1331/12**). This consideration of the specific implementation must, moreover, be from the point of view of the relevant skilled person under Art. 56 EPC, who may be identified on the basis of the invention's technical character. This was analogous to the approach of **T 928/03** (point 3.2 of the Reasons), which considered the actual contribution of each feature to the technical character by, for each feature, stripping away its non-technical content. Thus, "the extent to which the characterizing features contribute